Maximum Rank Reproducibility

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marr: Maximum Rank Reproducibility

marr (Maximum Rank Reproducibility) is a nonparametric approach that detects reproducible signals using a maximal rank statistic for high-dimensional biological data. In this R package, we implement functions that measures the reproducibility of features per sample pair and sample pairs per feature in high-dimensional biological replicate experiments. The user-friendly plot functions in this package also plot histograms of the reproducibility of features per sample pair and sample pairs per feature. Furthermore, our approach also allows the users to select optimal filtering threshold values for the identification of reproducible features and sample pairs based on output visualization checks (histograms). This package also provides the subset of data filtered by reproducible features and/or sample pairs. The R software is freely available from Bioconductor at `http://bioconductor.org/packages/marr`.

Method

Maximum Rank Reproducibility (MaRR) [1] was proposed to assess reproducibility in replicate experiments. We adapt this procedure in the context of metabolomics studies as it also involves technical or biological replicate experiments. The most significant advantage of MaRR is that it can be applied to different types of replicate experiments because it employs the rank scale.

We assume that each metabolite considered is associated with a continuous measure from each of the two replicate experiments. These measures can be a test statistic, p-value, q-value or fold change score. With M metabolites, we have two sets of measures: $x_1, \ldots, x_M$ and $y_1, \ldots, y_M$. Each metabolite m is associated with two ranks: $(R^x_m,\ R^y_m)$, where R^x_m is the rank of x_m and R^y_m is the rank of y_m. Figure 1 illustrates an example dataset of rank statistics for $M = 10,000$ metabolites under the ideal setting of a perfect split.

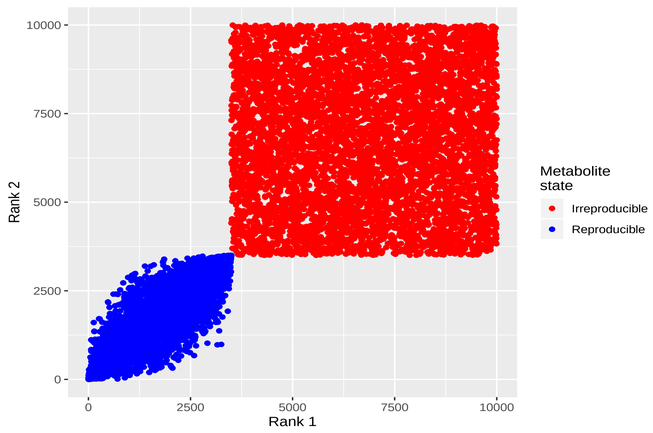


Fig. 1: Scatter plot showing rank pairs for 10,000 metabolites, 3,500 of them are reproducible. Blue points indicate reproducible metabolites, while red point indicate irreproducible metabolites.

Metabolites with reproducible measurements should be consistently highly ranked for both replicate experiments (blue, Figure 1), and are expected to have positive correlation in their ranks, whereas, metabolites with independent measures are assumed to have independent ranks (red, Figure 1).

Applications to Metabolomics Data

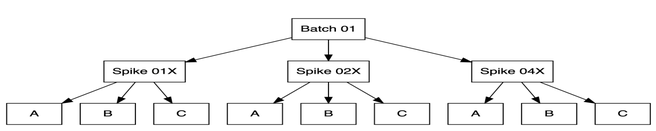
- Chronic Obstructive Pulmonary Disease (COPD) major cause of morbidity and mortality in US.
- Multi-center Genetic Epidemiology of COPD (COPDGene) studies the underlying genetic factors of COPD, [2].
- Visit 1 2008-2011 $n = 10,263$ individuals, aged 45-80 with $\geq$ 10 pack-year smoking history, no exacerbations for >30 days.
- Visit 2 2013-2017 $n = 6,758$ subjects returned, in-person 5-year visit (Visit 2)
- Each visit: spirometry before and after albuterol, quantitative CT imaging of the chest, blood sampling.

Data pre-processing

Pre-processing was performed using MSPrep R software [3]. The MSPrep pre-processing steps include i) Filtering: metabolites excluded if > 20% of samples were missing values, [4] , ii) imputation (Bayesian Principal Component Analysis (BPCA)) and iii) Normalization (quantile).

Technical(Tech) Data

Three technicians performed all steps of sample prep for a single human plasma sample (COPDGene visit 1). Six spiked-in control compounds at concentrations of 1X, 2X and 4X and two negative controls at 1X in all samples. Top layer indicates the Batch operators 01, 02, 03. The middle layer denotes the Spike-in 1x, 2x and 4x. The bottom layer indicates 3 technical replicates: A, B and C. After pre-processing, 860 metabolites remain. Figure 2 illustrates the hierarchical structure of the Tech data for Batch 01.



Biological-Technical(BioTech) Data

The data was analyzed using untargeted using Time of Flight (ToF) liquid chromatograph (LC) (Agilent 6210 Series) and Quadrupole ToF mass spectrometer (Agilent 6520). Data processing is described in [5] and available on Metabolomics Workbench (ST000601) `https://www.metabolomicsworkbench.org`. After pre-processing, 2860 metabolites remain. Figure 3 describes the data structure of BioTech data set.

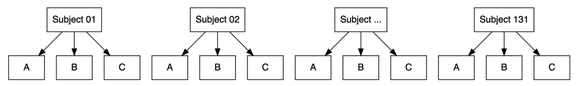


Fig. 3: Flow chart of the Biotech data set. The top layer indicates the subjects: Subject 1, Subject 2,..., Subject 131. The bottom layer under each subject indicates 3 technical replicates: A, B and C. In total, we have 393 technical replicates.

Biological (Bio) Data

The fresh frozen plasma collected at COPDGene Visit 2 ($n = 1,136$ subjects) was profiled with Metabolon Global Metabolomics platform using GC-MS and LC-MS metabolomic quantification protocol [6], [7]. After pre-processing 995 metabolites remain. Figure 4 illustrates the Bio data structure. Note: no technical replicates in this data set.

Fig. 4: Flow chart of 1130 subjects.

Summary Table of Reproducibility

Data set		Percentage of reproducible metabolites per sample pair			Percentage of reproducible sample pairs per metabolite		
		> 70%	> 80%	> 90%	> 70%	> 80%	> 90%
Tech	Operator layer	100.00	100.00	0.00	82.09	82.09	82.09
	Spike-in layer	88.89	11.11	0.00	81.28	81.05	80.23
	Replicate layer	100.00	92.59	0.00	85.23	84.30	83.02
BioTech	biological layer	49.85	9.59	0.73	64.44	60.80	55.07
	technical layer	90.08	0.76	0.25	68.29	63.78	59.09
Bio	biological layer	98.52	92.40	76.95	88.84	84.32	76.78

Table 1: Summary of reproducible metabolites per sample pair and reproducible sample pairs per metabolite for Tech, BioTech and Bio data sets.

Reproducibility of Tech Data

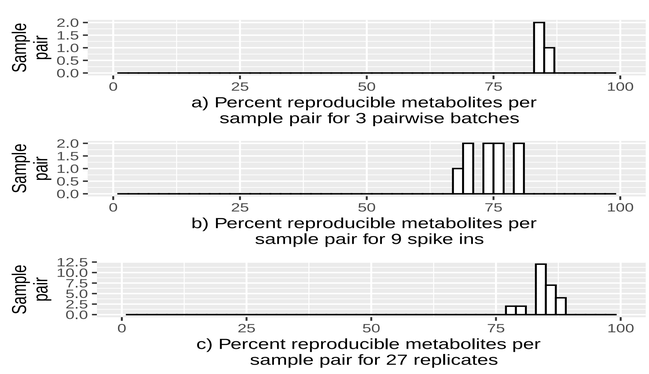


Fig. 5: Histograms showing reproducible metabolites for 3 layers. These histograms are in percent scale where the x-axes denote the percent scale of reproducible metabolites per sample pair. Reproducible metabolites a) Top layer (batches): 3 pairs of batches, b) Middle layer: (spike-ins): 9 pairs of spike-ins, c) Bottom layer: (technical replicates): 27 pairs of technical replicates.

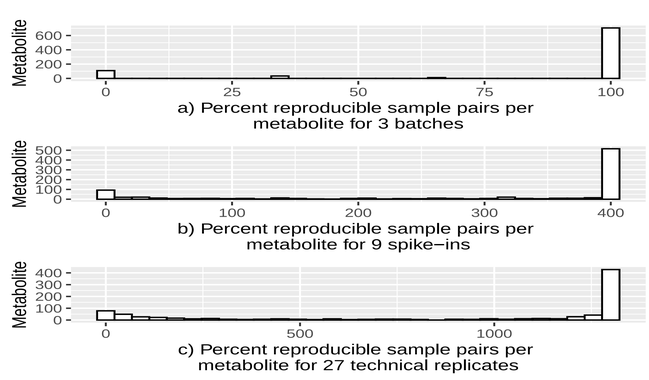


Fig. 6: Histograms showing reproducible sample pairs for 3 layers. These histograms are in percent scale where the x-axes denote the percent scale of reproducible sample pairs per metabolite. Reproducible metabolites a) Top layer (batches): 3 pairs of batches, b) Middle layer: (spike-ins): 9 pairs of spike-ins, c) Bottom layer: (technical replicates): 27 pairs of technical replicates.

In Figure 5, the higher percentage of reproducible metabolites per sample pair in the x-axis indicate stronger reproducibility between the two samples (sample pairs), e.g., the percent metabolites in the last right bin indicates that all or almost all the metabolites are reproducible for the number of sample pair combinations in the y-axis. The percentage of reproducible metabolites per sample pair (close to 100%) are much higher for Operator and technical layer compared to the spike-in layer data (Table 1). Figure 6 compares the percentage reproducible sample pairs per metabolite for all the 3 layers. Table 1 shows that the percentage of reproducible sample pairs per metabolite greater than different thresholds (70%, 80%, 90%) are more or less same, i.e., around 82%.

Reproducibility of Metabolites

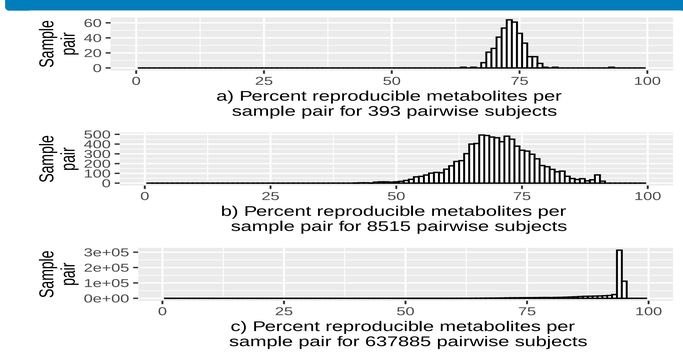


Fig. 7: Histograms showing reproducible metabolites per sample pair for BioTech and Bio data sets. These histograms are in percent scale where the x-axes denote the percent scale of reproducible metabolites. Reproducible metabolites a) 393 technical replicate sample pairs (BioTech), b) 8515 biological replicate sample pairs (BioTech), and c) 637885 biological replicate sample pairs (Bio).

Figure 7 illustrates that the reproducibility of metabolites per sample pair are higher for technical replicates of BioTech data set (Figure 7(a)) compared to biological replicates of the same data set (BioTech) (Figure 7 (b)). Table 1 also confirms the reproducibility of metabolites per sample pair is higher for technical replicates compared to its biological data set (BioTech). In BioTech data set, the percentage of reproducible metabolites per sample pair greater than 70% for technical replicates is 90.08% compared to 49.85% of biological replicates. In Figure 7 (b) and (c), we also report that the reproducibility of metabolites per sample pair is much higher for Bio data set compared the biological replicates of Biotech data set. However, there are differences between the two technologies for BioTech and Bio data sets since the samples were not collected on the same subjects at the same visit for a direct comparison.

Reproducibility of Sample Pairs

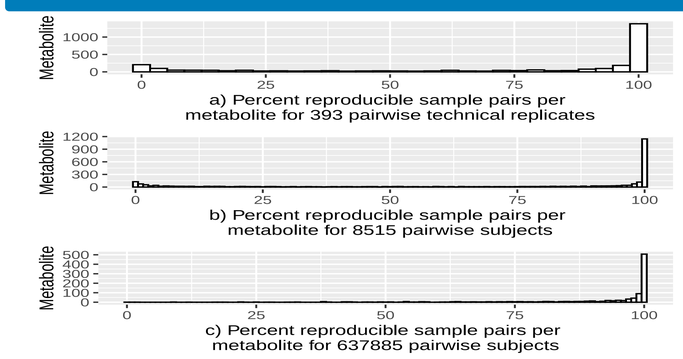


Fig. 8: Histograms showing reproducible sample pairs per metabolite for BioTech and Bio data sets. These histograms are in percent scale where the x-axes denote the percent scale of reproducible sample pairs. Reproducible sample pairs a) 393 technical replicate sample pairs (BioTech), b) 8515 biological replicate sample pairs (BioTech), and c) 637885 biological replicate sample pairs (Bio).

Figure 8 compares the percentage reproducible sample pairs per metabolite either across technical replicates (Figure 8(a)) or biological replicates (Figure 8(b),(c)). The percentage of reproducible sample pairs per metabolite across all combinations of biological sample pairs for Bio data set (Figure 8(c)) is higher than the technical replicates (Figure 8(a)) and biological replicates (Figure 8(b)) of the BioTech data set (Table 1). However, as mentioned above, the subjects and visit are different between BioTech and Bio data sets.

Concluding Remarks

We present MaRR as a quantitative approach for assessing the reproducibility of metabolites and sample pairs. This method helps us in detecting reproducible metabolites per sample pair as well as reproducible sample pairs per metabolite based on abundance measurements. The MaRR framework can also be used as a method for comparing reproducibility for different processing methods and technologies.

Acknowledgements

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References

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